

Title:

Higher-Dimensional Signal Projection Hypothesis (HDSPH)

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Abstract

The Higher-Dimensional Signal Projection Hypothesis (HDSPH) frames transient astrophysical signals as **noisy, operator-induced projections** of a latent stochastic field rather than direct emissions from uniquely identifiable sources. HDSPH replaces metaphysical appeals to extra spatial dimensions with a mathematically precise inverse-problem formulation: observations follow

$$[S_{\text{obs}} = \mathcal{H}[M] + \eta]$$

where (M) is a latent random field, $(\mathcal{H})$ a measurement operator (kernel, sampling, selection), and (η) observational noise. The framework quantifies recoverable information via mutual information $(I(M; O))$, models events as point processes driven by a latent intensity $(\lambda = \mathcal{G}(M))$, and predicts operator-dependent signatures such as convolutional distortion, event fragmentation, heavy-tailed fluence distributions, and cross-instrument anisotropies. HDSPH is explicitly falsifiable: it yields concrete statistical tests (overdispersion, tail-index estimation, mutual-information bounds, cross-correlation and entropy anomalies) and prescribes observational programs to discriminate projection-induced explanations from conventional emission models. The hypothesis is retained only when it demonstrably improves predictive performance or interpretability relative to simpler null models; otherwise, it reduces to a descriptive reparameterization of known phenomena.

*****Inspiration*****

*The author acknowledges the profound influence of **Maryam Mirzakhani's** work on the dynamics of moduli spaces and the geometry of Riemann surfaces, which indirectly shaped the stochastic inverse-problem perspective developed in this hypothesis.*

Introduction

Transient radio and time-domain phenomena present recurring interpretive challenges: isolated detections, non-repeating bursts, and heterogeneous populations resist unique source attribution when viewed through limited instruments. HDSPH addresses this by shifting the explanatory burden from hypothesizing new emitter classes to modeling **how** latent complexity is mapped into observation space. The central claim is not that new physics is required, but that **measurement geometry, instrument response, sampling, and noise** can produce apparent anomalies even when the underlying dynamics are structured but conventional.

The HDSPH program formalizes three core elements. First, a **latent stochastic field** (M) defined on a probability space encodes the true degrees of freedom; second, a **measurement operator**

$$\left(\mathcal{H}: L^2(\mathcal{M}) \rightarrow L^2(\mathcal{O})\right)$$

(often representable by a kernel $K(y, x)$) models convolution, selection, and sampling; third, an **event model** maps (M) to a latent intensity

$$\left(\lambda(x, t) = \mathcal{G}(M)\right)$$

that drives point processes observed as bursts or transients. Together these elements produce testable, operator-dependent predictions about population statistics, morphology, and cross-instrument coherence.

This manuscript develops HDSPH across four layers: foundational definitions and information-theoretic framing; a theoretical model of operators, information density, and message structure; a mathematical formalization yielding predictive statistics and inference recipes; and an applications chapter mapping the framework onto real problems (WOW! signal, FRBs, SETI, unidentified transients). Each layer emphasizes **operational falsifiability**: HDSPH specifies the statistics and observational programs required to prefer projection-based explanations over conventional emission models, and it prescribes regularization and model-selection strategies to address non-identifiability. The remainder of the paper details these components, provides canonical latent model classes and inference tools, and outlines concrete observational tests to evaluate the hypothesis.

HDSPH-0 Foundations

0.1 Projection Geometry

Definition

Projection geometry in HDSPH models the observation process as a measurement operator acting on a space of stochastic fields rather than as a purely geometric map:

$$[\mathcal{H}: \mathcal{F}(\mathcal{M}) \rightarrow \mathcal{F}(\mathcal{O})]$$

Where

$$(\mathcal{M})$$

is the latent state space and

$$(\mathcal{O})$$

is the observation space. Here $(\mathcal{F}(\cdot))$ denotes a specified function class (for example (L^2) or a space of square-integrable random fields). The operator $(\mathcal{H})$ represents instrument response, propagation, selection, and sampling.

Key clarification

$(\mathcal{M})$ is a statistical state space with potentially many degrees of freedom (possibly infinite); it represents latent variables and modes that are not directly accessible to observation.

Assumptions

A1

The physical system is described by a latent stochastic field (M) defined on a probability space $((\Omega, \mathcal{F}, P))$; (M) may contain unobservable degrees of freedom.

A2

Observation reduces accessible information. Formally, mutual information and entropies are related under a common measure: $[I(M; O) \leq H(O) \leq H(M),]$ with all information quantities computed with respect to the same underlying probability measure.

A3

The measurement operator is generally non-injective:

$$[\exists, M_1 \neq M_2 \quad \text{such that} \quad \mathcal{H}[M_1] = \mathcal{H}[M_2],] \text{equivalently} (\ker \mathcal{H} \neq 0).$$

A4

Information loss is a consequence of the structure of $(\mathcal{H})$ (null space, smoothing, sampling), not of physical destruction of information.

A5

Apparent simplicity in observations does not imply simplicity of the latent structure; compression by $(\mathcal{H})$ can hide complexity.

Mathematical Formulation

Basic relation (shorthand): $[O = \mathcal{H}[M] + \eta]$ with (η) an observation noise term. More precisely, $[\mathcal{H}: L^2(\mathcal{M}) \rightarrow L^2(\mathcal{O}),]$ and a common kernel representation is

$$\left[(\mathcal{H}M)(y) = \int_{\mathcal{M}} K(y, x) , M(x), d\mu(x), \right]$$

where $(K(y, x))$ encodes instrument response, propagation, and selection effects.

Information loss

Replace informal difference with a mutual-information-based loss:

$$[L_{\mathcal{H}}; =; I(M) - I(M; O); =; I(M \mid O).]$$

This quantity measures irrecoverable latent information given the observation and the assumed prior on (M) .

Observable information fraction

When $(I(M))$ is defined,

$$\left[R = \frac{I(M; O)}{I(M)}, \quad 0 \leq R \leq 1, \right]$$

with the caveat that both numerator and denominator must be defined under the same measure and model class.

Physical Interpretation

Projection is operationally decomposed into selection, convolution with an instrument kernel, finite sampling, and additive noise. Observed signals are compressed, operator-dependent realizations of a latent stochastic field.

Observable Consequences

C1 Apparently isolated events

Single detections can be sparse samples of localized peaks in the latent intensity.

C2 Incomplete information recovery

Reconstruction is limited by ill-posed inversion (null space and instability), not only by data scarcity.

C3 Degeneracy of interpretation

Non-injectivity yields multiple latent explanations consistent with the same observation.

C4 Anomalous temporal behavior

Non-stationarity in $(M(x,t))$ produces non-standard recurrence patterns.

C5 Non-intuitive signal morphology

Observed shapes are distorted by convolution with the measurement kernel and by sampling.

Limitations

L1 Hidden variables in $(\mathcal{M})$ are model assumptions and not directly observed.

L2 The operator $(\mathcal{H})$ must be empirically inferred as part of an inverse problem.

L3 No assumption is made about the physical nature of latent variables.

L4 The model does not imply intelligence or intentional signal generation.

L5 Falsifiability is operationalized through statistical tests on ensembles of observations and explicit null hypotheses.

Foundational Statement

Every observation (O) is modeled as the result of a measurement operator acting on a latent stochastic field plus noise: $[O = \mathcal{H}[M] + \eta.]$ Observed structures are therefore compressed, operator-dependent realizations of (M); recoverable information is quantified by $(I(M; O))$ and irrecoverable modes lie in $(\ker \mathcal{H})$.

0.2 Information Theory

Definition

The latent field (M) is a random element with law

$$(P_M): [M: (\Omega, \mathcal{F}, P) \rightarrow L^2(\mathcal{M}), \quad M \sim P_M.]$$

Signals are realizations of stochastic processes.

Mathematical Formulation

Shannon entropy (continuous form)

For a random variable (X) with density (p_X) relative to a dominating measure

$$(\mu), \left[H(X) = - \int p_X(x) \log p_X(x), d\mu(x), \right]$$

with the usual caveats for differential entropy.

Observable information and mutual information

Observable information is characterized by $(H(O))$ and, crucially, by mutual information:

$$I(M; O) = H(M) - H(M | O)$$

which quantifies the portion of latent information recoverable from observations.

Physical Interpretation

Apparent randomness in (O) can arise from increased conditional entropy ($\mathcal{H}(M | O)$) induced by ($\mathcal{H}$) and noise, not necessarily from intrinsic randomness of the source.

Observable Consequences

Hidden structures may persist under low mutual information channels.

Entropy estimates can underestimate latent complexity unless priors and measure choices are explicit.

Multiple observations and appropriate priors are generally required for meaningful reconstruction.

Limitations

Entropy and mutual information depend on the chosen probability measure and estimator.

Full recovery requires prior assumptions (Bayesian dependence).

Complete reconstruction is generally non-identifiable without additional constraints.

0.3 Extra Dimensions

Definition

Model the latent state space as a product of observable and latent coordinates:

$$[\mathcal{M} = \mathcal{X} \times \mathcal{Z},]$$

Where

$$(\mathcal{X})$$

are observable coordinates and

$$(\mathcal{Z})$$

are latent coordinates (possibly high- or infinite-dimensional).

Key Clarification

($\mathcal{Z}$) denotes hidden degrees of freedom in state space, not additional physical space-time dimensions.

Mathematical Formulation

Treat($\mathcal{Z}$) as a high-dimensional or separable Hilbert space when convenient; write $(\dim(\mathcal{Z}))$ to indicate degrees of freedom rather than physical dimensions.

0.4 Observer Constraints

Definition

The observer (detector system) is modeled as a measurement system defined by a pair:

$$D = (\mathcal{H}, \Sigma)$$

where:

- $\mathcal{H}$: measurement operator (instrument response, propagation, selection, sampling)
- Σ : noise covariance on the observation space

Assumptions

D1

No observer can access the complete latent informational manifold; observation is fundamentally limited by the instrument.

D2

Detection systems possess finite resolution, characterized by the kernel $K(y,x)$ and the noise covariance Σ .

D3

Observation itself restricts accessible information; different instruments

$$(\mathcal{H}^1 \neq \mathcal{H}^2)$$

produce different projections.

D4

The observed signal depends on detector capabilities, not only on the latent field.

Mathematical Formulation

Observation model:

$$O = \mathcal{H}[M] + \eta, \eta \sim N(0, \Sigma)$$

(with the option to replace Gaussian noise by an alternative distribution when appropriate).

Corrected information constraint:

Under $\mathcal{H}$,

$$I(M; O) \leq \mathcal{H}(O) \leq \mathcal{H}(M)$$

and the null space $\ker \mathcal{H}$ identifies latent modes that are invisible to the observer.

Total observable information hierarchy:

$$I_{\text{obs}} \leq I_{\text{proj}} \leq I_{\text{latent}}$$

where:

- I_{latent} : information content of the latent field M
- I_{proj} : information remaining after projection through $\mathcal{H}$ (before noise)
- I_{obs} : information actually recovered after noise and finite sampling

Physical Interpretation

Observational limitations arise from:

- finite instrument resolution (discretization, band limits)
- stochastic noise (thermal, instrumental, environmental)
- non-invertibility of the measurement operator (null space, smoothing, convolution)

Even if a projection contains significant information, detector limitations may prevent full recovery. Therefore, the observed signal depends on:

1. the latent structure M
2. the projection/measurement operator $\mathcal{H}$
3. the observer's instrument and noise characteristics Σ

Observable Consequences

- **Detector-dependent signal variation:** different instruments ($\mathcal{H}_1$ vs $\mathcal{H}_2$) may produce systematically different signal morphologies from the same latent source.
- **Missing information channels:** latent modes lying in $\ker \mathcal{H}$ are fundamentally irrecoverable with a given instrument.
- **Incomplete signal reconstruction:** posterior uncertainty remains broad even after optimal inversion.
- **Sensitivity-dependent event detection:** detection rate Γ scales with instrument sensitivity and windowing function.

Limitations

- Instrument model dependence: separation of instrument effects from latent structure is non-trivial and requires careful calibration.
- Some observational effects are fundamentally indistinguishable without additional constraints or multi-instrument data.
- Measurement effects (noise, calibration errors) may mimic or obscure projection-induced signatures.

Relationship to Foundational Statement

Incorporating observer constraints into the HDSPH framework modifies the foundational statement:

$$O = \mathcal{H}[M] + \eta$$

Every observation is the result of applying a specific measurement operator $\mathcal{H}$ to a latent stochastic field M , contaminated by instrument noise η . Different observers (different $\mathcal{H}$ and Σ) will, in general, obtain different projections of the same underlying latent structure. Thus, observed structures are **instrument-conditional compressed realizations** of M .

Summary

*This foundational layer defines HDSPH as a **stochastic inverse-problem framework** in which observed signals are noisy, compressed projections of a latent stochastic field. It replaces metaphysical language about “higher dimensions” with mathematically defined latent state spaces, measurement operators ($\mathcal{H}$), explicit noise models, and operational tests for falsifiability and identifiability.*

HDSPH-1 Theoretical Model

Purpose

The purpose of this chapter is to establish the internal mathematical framework of HDSPH. While HDSPH-0 introduced foundational assumptions regarding projection, information, dimensionality, and observational constraints, HDSPH-1 defines the mathematical objects required to describe latent stochastic processes and their observed manifestations.

1.1 Projection Operator

Definition

The projection operator is the measurement operator acting on a latent stochastic field:

$$[\mathcal{H}: \mathcal{F}(\mathcal{M}) \rightarrow \mathcal{F}(\mathcal{O}).]$$

The operator maps a latent field into an observable signal space. It induces convolution, noise mixing, smoothing, and loss of identifiability.

Assumptions

P1

A latent stochastic structure (M) exists on a probability space $((\Omega, \mathcal{F}, P))$.

P2

Observations yield transformed realizations of (M) through $(\mathcal{H})$.

P3

The operator $(\mathcal{H})$ is generally non-invertible; a unique inverse does not exist in general.

P4

Information may be preserved in latent form but irrecoverable in practice due to ill-posed inversion.

Mathematical Formulation

Fundamental relation $[O = \mathcal{H}[M] + \eta,]$ where (M) is the latent stochastic field, $(\mathcal{H})$ the measurement operator, (O) the observed signal, and (η) additive observation noise.

Operator domain and codomain $[\mathcal{H}: L^2(\mathcal{M}) \rightarrow L^2(\mathcal{O}).]$

Kernel representation $[(\mathcal{H}M)(y) = \int_{\mathcal{M}} K(y, x), M(x), d\mu(x),]$ where $(K(y, x))$ encodes instrument response, propagation, and selection effects.

Projection loss

Use a mutual information based loss: $[L_{\mathcal{H}} = I(M) - I(M; O) = I(M | O)].$ Loss arises from the null space of $(\mathcal{H})$ and from noise and smoothing induced by the kernel.

Physical Interpretation

The projection operator is operationally equivalent to convolution with an instrument response, dimensional reduction into observation coordinates, stochastic sampling of a latent field, and additive noise contamination.

Observable Consequences

Signal ambiguity due to non-injectivity of $(\mathcal{H})$.

Partial recoverability of latent structure.

Apparent randomness induced by projection and noise.

Non-unique inverse solutions and model degeneracy.

Limitations

$(\mathcal{H})$ must be empirically inferred from data.

The inverse problem is ill-posed and unstable without regularization.

Multiple latent fields may explain identical observations.

1.2 Informational Manifold**Definition**

Replace geometric manifold language with a probability space formulation: $[(\Omega, \mathcal{F}, P)]$ with latent field $[M: \Omega \rightarrow R]$ interpreted as a random field or stochastic process.

Assumptions**M1**

Information is encoded as a stochastic field over the probability space.

M2

Observed messages correspond to localized realizations of this field.

M3

Observation corresponds to partial sampling of the underlying probability structure.

Mathematical Formulation

Local information density $\left[\rho_I(x) = \frac{dI}{d\mu(x)}\right]$ defined with respect to a chosen measure $(\mu)on(\mathcal{M})$.

Total information $[I(M) = \int_{\mathcal{M}} \rho_I(x), d\mu(x).]$ Integrals are taken over the measure space rather than a purely geometric domain.

Physical Interpretation

A message is a trajectory or localized high-probability region in the stochastic field, a coherent structure in latent state space, and a spatio-temporal realization of the random process.

Observable Consequences

Fragmented signal realizations.

Multiple valid reconstructions consistent with the same observations.

Hidden correlations and structured dependencies in the latent field.

Limitations

Metric and topological structure of $(\mathcal{M})$ are not directly observed and must be inferred.

Only statistical properties of (M) are accessible from data.

1.3 Event Distribution

Definition

Events are realizations of a stochastic point process driven by a latent intensity field:

$$[\lambda(x, t) = f(M(x, t)).]$$

Assumptions

E1

Events arise from threshold crossings or intensity realizations of $(\lambda(x, t))$.

E2

Event occurrence is sparse in latent space.

E3

Event statistics depend on the structure and non-stationarity of the latent field.

Mathematical Formulation

Observation process

$$N(A) \sim \text{Poisson} \left(\int_A \lambda(x, t) d\mu(x) dt \right)$$

Detection condition

Detection corresponds to support overlap between the latent intensity and the observation window: $[\text{supp}(\lambda) \cap W \neq \emptyset]$

Physical Interpretation

Observed bursts correspond to sparse realizations of a Poisson process, localized peaks in the latent intensity field, and stochastic sampling of a non-stationary process.

Observable Consequences

Non-periodic event structure.

Burst clustering and variable recurrence rates.

Heavy-tailed event amplitude or fluence distributions.

Limitations

The true intensity field (λ) is not directly observable.

Event rates are inferred and depend on detector sensitivity and windowing.

Different latent fields can produce identical event statistics.

1.4 Message Structure

Definition

A message is a structured realization of a latent stochastic field projected into observation space:

$$[S_{\text{obs}} = \mathcal{H}[S_{\text{latent}}].]$$

Assumptions**S1**

Messages correspond to structured regions in latent state space.

S2

Observed signals are partial projections of these structures.

S3

Latent complexity typically exceeds observed complexity.

Mathematical Formulation**Latent message**

$$[S_{\text{latent}} \in L^2(\mathcal{M}).]$$

Observed message

$$[S_{\text{obs}} = \mathcal{H}[S_{\text{latent}}].]$$

Recoverability ratio

Use mutual information when possible: $\left[R = \frac{I(S_{\text{latent}}; S_{\text{obs}})}{I(S_{\text{latent}})} \right]$

Physical Interpretation

A transient radio signal may be a cross-section of a latent field, a partial projection of a coherent structure, or a temporally localized sampling event.

Observable Consequences

Incomplete signal reconstruction from single observations.

Hidden multi-layer structure and ambiguous source interpretation.

Irreducible uncertainty in inversion without priors or multiple observations.

Limitations

Full reconstruction from a single observation is generally non-identifiable.

Inverse solutions require priors or regularization.

Latent structure cannot be uniquely recovered without additional constraints.

Chapter Summary

HDSPH-1 defines four core mathematical objects:

Measurement operator ($\mathcal{H}$)

Latent stochastic field (M)

Poisson event process (N)

Message structure as projected realizations

Together they form a stochastic inverse-problem framework for interpreting transient astrophysical signals under partial observability.

Short additions to support inference and testing

Identifiability and regularization

Identifiability holds only up to $(\ker \mathcal{H})$. Practical inversion requires regularization or priors.

Canonical choices include Tikhonov regularization, sparsity priors, or Bayesian priors $(\pi(M))$.

Posterior distributions $(p(M | O) \propto p(O | M)\pi(M))$ quantify remaining uncertainty and the effective irrecoverable information.

Canonical latent model classes

State explicit model classes to be used in later chapters: (1) Gaussian random fields with Matérn

covariance; (2) heavy-tailed (α)-stable fields; (3) non-stationary Gaussian processes. Each class implies distinct projection signatures and testable statistics.

Operational falsifiability checklist

Convert conceptual falsifiability into concrete tests: (i) overdispersion and goodness-of-fit tests against homogeneous Poisson; (ii) tail-index estimation for heavy tails; (iii) mutual information lower-bound estimation; (iv) cross-instrument coherence and Ripley-type spatial clustering tests.

HDSPH-2 Mathematical Formulation Revised Scientific Version

Introduction

The purpose of this chapter is to formalize HDSPH in a mathematically consistent framework. The equations presented here are not derived from a fundamental physical theory but are formulated as a stochastic inverse problem with operator-induced observation constraints, intended to produce falsifiable statistical predictions.

2.1 Projection Equations

Definition

Projection equations describe the mapping from a latent stochastic field to an observed signal through a measurement operator:

$$[\mathcal{H}: L^2(\mathcal{M}) \rightarrow L^2(\mathcal{O}).]$$

Assumptions

PE1

The system is defined on a latent space with higher degrees of freedom than the observation space; $(\mathcal{M})$ may be high- or infinite-dimensional.

PE2

Observation is modeled as the application of a linear or nonlinear measurement operator $(\mathcal{H})$.

PE3

The operator $(\mathcal{H})$ is generally non-invertible, leading to loss of identifiability and a nontrivial null space.

Mathematical Formulation

Fundamental equation

$[O = \mathcal{H}[M] + \eta,]$ where (M) is the latent stochastic field, (O) the observed signal, and (η) an additive noise term (noise model specified by the observer).

Continuous field formulation

$[O(x, t) = \mathcal{H}[M(x_1, x_2, \dots, x_n, t)],]$ emphasizing that the observable coordinates (x, t) are lower dimensional projections or summaries of the latent coordinates.

Kernel representation (linear case)

$[(\mathcal{H}M)(y) = \int_{\mathcal{M}} K(y, x), M(x), d\mu(x),]$ where $(K(y, x))$ is the measurement kernel encoding instrument response, propagation, selection effects, and (μ) is the measure on $(\mathcal{M})$. For nonlinear $(\mathcal{H})$, write $(\mathcal{H}[M] = \mathcal{G}(K \star M))$ or an equivalent operator decomposition.

Physical Interpretation

The kernel ($K(y,x)$) encodes instrument response, propagation effects, and projection-induced mixing of degrees of freedom. Observation acts as convolution/smoothing in latent coordinates combined with sampling and noise, rather than as a simple geometric projection.

Observable Consequences

Non-unique inverse solutions due to $(\ker \mathcal{H})$.

Structural degeneracy in source reconstruction.

Signal distortion induced by operator smoothing and null modes.

Compression of latent structure into lower-dimensional observables.

Limitations

The kernel ($K(y,x)$) is not directly measurable and must be inferred.

The inverse problem is ill-posed and requires regularization or priors.

Multiple latent fields may generate identical observables under $(\mathcal{H})$.

2.2 Information Recovery

Definition

Information recovery quantifies the mutual information between latent and observed processes under the measurement operator $(\mathcal{H})$.

Assumptions

IR1

The operator $(\mathcal{H})$ reduces accessible information relative to the latent field.

IR2

Recovery is limited by the null space and smoothing properties of $(\mathcal{H})$.

IR3

Observed information provides a projection-dependent lower bound on latent information.

Mathematical Formulation

Recovery fraction

$\left[R = \frac{I(M;O)}{I(M)}, \right]$ where $(I(M;O))$ is mutual information and $(I(M))$ denotes the information content of the latent field under the chosen prior/model class.

Information loss

$$[L = I(M) - I(M;O) = I(M | O).]$$

Relative loss

$$[\eta = 1 - R.]$$

Mutual information identity

$[I(M; O) = H(M) - H(M | O),]$ with entropies and conditional entropies defined with respect to the same underlying probability measure.

Physical Interpretation

Low (R) indicates a high-dimensional latent structure, strong compression by the measurement operator, and high uncertainty in inverse reconstruction.

Observable Consequences

Apparent randomness in projections even for structured latent fields.

Non-unique reconstructions and broad posterior distributions.

Increased empirical entropy of observables relative to recoverable latent structure.

Limitations

True $I(M)$ is not directly measurable and depends on the chosen prior/model class.

Recovery estimates require explicit priors or model families and are sensitive to model misspecification.

Mutual information estimation in high dimensions is challenging and requires careful lower-bound techniques or surrogate statistics.

2.3 Intersection Statistics**Definition**

Event occurrence is modeled as a stochastic point process driven by a latent intensity field $(\lambda(x, t))$.

Assumptions**IS1**

Events are generated by a latent intensity field $(\lambda(x, t))$.

IS2

Event occurrence is stochastic and may be non-stationary.

IS3

Observation corresponds to sampling of a point process through the measurement operator and observation window.

Mathematical Formulation**Latent intensity field**

$[\lambda(x, t) \geq 0, \quad \lambda = f(M(x, t)),]$ where (f) is a link function (e.g., thresholding, exponential link, or other nonlinear mapping).

Observation process

$$\left[N(A) \sim \text{Poisson} \left(\int_A \lambda(x, t) d\mu(x) dt \right) \right]$$

for observation region (A) in latent coordinates mapped into the observation window.

Detection condition (operational)

$$\left[\int_A \lambda(x, t) d\mu(x), dt > 0 \right]$$

or, in practice, exceedance of a detection threshold after application of ($\mathcal{H}$) and noise.

Expected detections

$$[\langle k \rangle = \int_A \lambda(x, t) d\mu(x), dt]$$

with detector efficiency and windowing factors applied as needed.

Physical Interpretation

Observed bursts correspond to sparse realizations of a Poisson (or Cox) process driven by localized activation of the latent intensity field; non-stationarity and clustering arise naturally from structure in $(\lambda(x, t))$.

Observable Consequences

Burst-like event distributions and clustering in time and space.

Non-periodic occurrence patterns and heavy-tailed interarrival statistics.

Apparent heterogeneity in event rates across observation windows.

Limitations

The true intensity field (λ) is latent and must be inferred.

Estimation depends on detector sensitivity, windowing, and selection effects.

Different latent fields can produce statistically similar Poisson observations; hierarchical or marked point process models may be required to disambiguate.

2.4 Information Density Functions

Definition

Information density is defined as a measure density over the latent space: $\left[\rho_I(x) = \frac{dI}{d\mu(x)}, \right]$ with (μ) the chosen measure on $(\mathcal{M})$.

Assumptions

ID1

Information is non-uniformly distributed across latent coordinates.

ID2

Localized high-density structures dominate observable signals.

ID3

Detection probability increases with local information density.

Mathematical Formulation**Total information**

$$\left[I(M) = \int_{\mathcal{M}} \rho_I(x), d\mu(x). \right]$$

Projected information density

$[\rho_{\text{obs}}(y) = \int_{\mathcal{M}} K(y, x), \rho_I(x), d\mu(x),]$ which describes how latent information is redistributed into observation coordinates by the kernel.

Density contrast

$\left[\delta_I = \frac{\rho_I - \bar{\rho}_I}{\bar{\rho}_I}, \right]$ where $(\bar{\rho}_I)$ is the mean information density under the chosen measure.

Physical Interpretation

High information-density regions act as high-probability emission zones and dominate observable structure; they function as attractors in measurement space after projection by $(\mathcal{H})$.

Observable Consequences

Clustered event distributions and uneven detection rates.

Heavy-tailed signal populations driven by localized high-density regions.

Sensitivity of observed statistics to the overlap between (ρ_I) and the observation window.

Limitations

(ρ_I) is latent and must be inferred via inverse methods.

Inference is sensitive to model assumptions and regularization choices.

Projected density estimates require careful deconvolution or forward modeling.

2.5 Predictive Models**Definition**

Predictive models describe observable statistics derived from the latent stochastic field under the measurement operator and observation protocol.

Assumptions**PM1**

Observed statistics are operator-dependent and sensitive to instrument response.

PM2

Latent structure leaves statistical signatures in observed data that can be modeled and tested.

PM3

Models must be falsifiable through explicit statistical testing against null hypotheses.

Mathematical Formulation

Detection rate

$[\Gamma = \alpha \int_{\mathcal{M}} \lambda(x), W(x), d\mu(x),]$ where (Γ) is the expected detection rate, (α) detector efficiency, $(W(x))$ the observation window function, and $(\lambda(x))$ the latent intensity.

Complexity scaling

$[C_{\text{obs}} = R \cdot C_{\text{latent}},]$ where (C) denotes an appropriate complexity measure (e.g., effective dimension, entropy) and (R) the recoverability fraction.

Projection variance

$[V_{\text{obs}} \propto \sigma_{\mathcal{H}}^2,]$ indicating that observed variance scales with operator-induced smoothing and noise characteristics.

Physical Interpretation

Observed statistics are jointly determined by latent field structure, measurement operator properties, and sampling/windowing effects; predictive models must combine forward operator modeling with stochastic latent priors.

Observable Consequences

Non-Gaussian signal populations and rare high-amplitude events.

Structured noise-like distributions and hidden correlations across events.

Sensitivity of detection statistics to survey strategy and instrument configuration.

Limitations

Predictions depend on accurate operator modeling and prior specification.

The latent field is not directly observable; validation requires large, well-characterized datasets.

Model selection and comparison require robust statistical criteria (e.g., likelihood ratios, Bayesian evidence, predictive checks).

Chapter Summary

HDSPH-2 formalizes the framework as a stochastic inverse problem defined by a latent intensity field, a measurement operator, and a point process for events. It establishes five core components:

Measurement operator ($\mathcal{H}$)

Latent field (M)

Poisson (or Cox) event process (N)

Information recovery quantified via mutual information

Statistical prediction under projection constraints

Together these elements provide a mathematically consistent basis for deriving testable predictions and for designing inference procedures that account for operator effects, latent complexity, and observational selection.

Short operational additions for methods and testing

Canonical link functions and generative choices

Specify candidate link functions (f) mapping (M) to (λ): (i) exponential link ($\lambda = \exp(g(M))$) for positive intensity; (ii) thresholded activation ($\lambda = \mathbf{1}_{g(M) > \theta}$) for event triggering; (iii) Cox process formulation with (λ) itself a stochastic field. Each choice implies distinct likelihoods and inference strategies.

Regularization and identifiability

State canonical regularizers: Tikhonov ($(\|LM\|_2^2)$), sparsity priors ((l_1)), and hierarchical Bayesian priors. Quantify identifiability in terms of ($\ker \mathcal{H}$) and posterior concentration; use posterior predictive checks to assess practical identifiability.

Model validation checklist

Provide concrete tests: (i) overdispersion and goodness-of-fit against homogeneous Poisson; (ii) tail-index estimation for heavy tails (Hill estimator or Bayesian tail models); (iii) mutual information lower bounds (variational estimators); (iv) Ripley's K and cross-correlation tests for clustering; (v) cross-instrument consistency checks to separate instrument effects from latent structure.

HDSPH-3 Applications

Introduction

The purpose of this chapter is to evaluate how the HDSPH framework maps onto empirical transient astrophysical datasets. These case studies are **not** presented as proof of HDSPH; they define testable observational regimes in which HDSPH-derived statistical models can be compared against conventional emission-based explanations. The central question is whether projection-induced stochastic models improve explanatory power for transient signal populations.

3.1 WOW! Signal

Observational Background

The WOW! Signal (1977) is a single narrow-band radio transient characterized by high signal-to-noise ratio, narrow spectral bandwidth, duration (≈ 72) s, and non-repetition in follow-up observations.

HDSPH Model Mapping

Model the observation as a single realization of a point process conditioned on a latent field: $[S_{\text{obs}}(t) = \mathcal{H}(M(x, t)) + \eta(t),]$ with a detection defined operationally by exceedance of a threshold after projection and noise: $[P(S_{\text{obs}} > \theta) > 0.]$ Equivalently, the event can be expressed as a rare exceedance of the latent intensity: $[\int_A \lambda(x, t), d\mu(x), dt \geq \theta,]$ with $(\lambda = f(M))$ a chosen link from latent field to intensity.

Interpretation

Under HDSPH the WOW! Signal is interpreted as a sparse, high-amplitude realization of a latent stochastic field — a single draw from a heavy-tailed event distribution rather than necessarily a unique, persistent astrophysical transmitter.

Predicted Statistical Features

High signal-to-background ratio for rare latent peaks.

Low recurrence probability consistent with heavy-tailed extremes.

Non-Gaussian amplitude distribution (power-law or subexponential tails).

Strong sensitivity to observation window and instrument kernel (different $(\mathcal{H})$ produce different detectabilities).

Falsification Criteria (operational)

Show that narrow-band transients are fully explained by known astrophysical emitters with predictive models that outperform HDSPH in likelihood or predictive checks.

Demonstrate absence of heavy-tailed behavior in large, well-characterized surveys (tail-index tests, Hill estimator, or Bayesian tail models).

Establish recurrence statistics consistent with a homogeneous Poisson emitter after accounting for selection effects (goodness-of-fit and overdispersion tests).

3.2 Fast Radio Bursts (FRBs)

Observational Background

FRBs are millisecond transients with extremely high apparent luminosity, diverse repetition patterns, and, in many cases, cosmological dispersion measures.

HDSPH Model Mapping

Model FRBs as realizations of a non-stationary point process driven by a latent intensity:

$$[N(t) \sim \text{Poisson}(\lambda(t)), \quad \lambda(t) = \mathcal{G}(M(x, t)),]$$

with observed signal

$$[S_{\text{obs}}(t) = \mathcal{H}(M(x, t)) + \eta(t).]$$

Interpretation

Diversity among FRBs (repeaters vs non-repeaters) can arise from different regimes of the latent intensity $(\lambda(x, t))$ and different non-stationarity patterns, rather than necessarily distinct emission mechanisms.

Predicted Features

Heavy-tailed fluence distribution across the population.

Temporal clustering for repeating sources (overdispersion relative to homogeneous Poisson).

Weak or complex correlation between observed fluence and latent structural parameters after projection.

Statistical similarity across apparently distinct source classes when analyzed in projected feature space.

Falsification Criteria

Clear separation of FRB populations into distinct astrophysical mechanisms with distinct predictive models that HDSPH cannot reproduce.

Lack of shared latent-structure signatures across populations (mutual information and clustering tests).

Intensity distributions demonstrably Gaussian and stationary after correcting for selection and propagation effects.

3.3 SETI

Observational Background

SETI searches typically assume artificial transmitters with periodic or encoded signals and persistent emission.

HDSPPH Reformulation

Treat SETI data as projected realizations of a latent structured field:

$$[S_{\text{obs}}(t, f) = \mathcal{H}(M(x, t, f)) + \eta,]$$

where (M) need not represent intentional communication.

Key Conceptual Shift

Replace “search for transmitters” with **search for statistically anomalous structure** in a projected stochastic field: look for deviations from instrument- and propagation-matched null models rather than only for narrowband periodicity.

Predicted Observable Features

Isolated high-significance events inconsistent with thermal noise.

Non-periodic structured noise with cross-frequency correlations after deconvolution by ($\mathcal{H}$).

Clustering inconsistent with white-noise models (spatial, temporal, or spectral).

Detection Rate Model

$$\left[\Gamma = \alpha \int_{\mathcal{M}} \lambda(x), W(x), d\mu(x), \right]$$

with (α) detector efficiency and (W) the survey window.

Falsification Criteria

All candidate signals are explained by known astrophysical or instrumental models with no residual anomalies after forward modeling of ($\mathcal{H}$).

No statistically significant deviation from white-noise or known noise models in large, well-calibrated datasets.

No clustering beyond Poisson baseline after accounting for selection effects.

3.4 Unidentified Transient Signals

Observational Background

Modern surveys report many weakly classified transients that are often reclassified later as known phenomena.

HDSPPH Model Mapping

Model unidentified transients as projected realizations:

$$[S_{\text{obs}} = \mathcal{H}[M] + \eta,]$$

with (M) containing structured but unresolved latent components.

Key Statistical Interpretation

Rather than invoking unknown source classes, treat these events as **tail events** of a structured stochastic field whose projection and sampling produce apparent novelty.

Predicted Features

Non-uniform spatial distribution tied to overlap of (ρ_l) with survey windows.

Weak clustering in time–frequency space reflecting latent structure and instrument response.

High variance in localization accuracy due to projection and kernel smoothing.

Strong dependence on instrument response function and survey strategy.

Falsification Criteria

Demonstrate that all transients are consistently reducible to known astrophysical categories with predictive models that outperform HDSPH.

Show no residual statistical anomalies remain after classification and forward modeling.

Show that latent-field modeling yields no improvement in predictive performance or interpretability.

Chapter Summary

HDSPH-3 maps the theoretical framework onto observational regimes by reformulating:

signals as stochastic realizations,

events as Poisson/Cox processes driven by latent intensity fields,

projection as the measurement operator ($\mathcal{H}$), and

anomalies as statistical deviations from baseline models.

Final Scientific Interpretation

HDSPH does **not** posit a new class of physical emitters. It provides a **statistical inverse-problem framework** for interpreting transient astrophysical signals under partial observability. The framework is operational: it yields explicit generative models, falsifiable predictions, and a concrete set of statistical tests (tail-index estimation, overdispersion, mutual information lower bounds, cross-instrument consistency, and posterior predictive checks) that can be applied to survey data to evaluate whether projection-induced explanations improve explanatory and predictive power relative to conventional emission-based models.

HDSPH-4 : Testable Predictions

Introduction

A scientific hypothesis must be evaluated through quantitative predictive performance against observational data.

This chapter defines HDSPH as a **statistical model class over transient astrophysical data**, and identifies measurable deviations from standard null hypotheses such as Poisson noise, Gaussian emission models, and stationary stochastic processes.

All predictions are derived from the operator form:

$$[S_{obs} = \mathcal{H}(M) + \eta]$$

and latent intensity formulation:

$$[\lambda(x, t) = \mathcal{G}(M(x, t))]$$

4.1 Observable Signatures

Definition

Observable signatures are statistical deviations in observed signal ensembles relative to null models of independent emission processes.

Prediction OS1: Information Deficit (Improved Formulation)

Instead of direct ratio:

$$\left[R = \frac{I(M; O)}{I(M)} < 1 \right]$$

Testable form:

$$[I(M; O) \ll H(O)]$$

Empirical implication:

inverse problem is ill-posed

multiple latent models achieve equivalent likelihood

posterior distributions remain broad under Bayesian inference

Prediction OS2: Projection-Induced Distortion

$$\left[S_{obs}(y) = \int K(y, x) M(x), dx \right]$$

Testable implication:

Observed signal morphology should exhibit:

convolution-induced smoothing

non-local mixing of spectral components

mismatch between temporal and spectral structure

Prediction OS3: Event Fragmentation

A single latent structure produces multiple observable events.

Statistical formulation:

$$[P(N_{cluster} > 1 | M) > P_{Poisson}]$$

Observable implication:

excess event clustering beyond Poisson baseline

cross-event correlation in feature space

Prediction OS4: Geometry-Dependent Visibility

Detection probability depends on overlap between latent support and observation window:

$$\left[P_{det} = \int \lambda(x, t), W(x, t), dx, dt \right]$$

Observable implication:

strong dependence of detection rate on survey strategy

anisotropic event visibility

variable repeatability across instruments

4.2 Statistical Predictions

Prediction SP1: Non-Uniform Event Rates

$$\left[\Gamma = \alpha \int \lambda(x), W(x), d\mu(x) \right]$$

Testable implication:

spatial clustering in event catalogs

deviation from homogeneous Poisson process

Prediction SP2: Heavy-Tail Distributions

Formal statement:

$$[P(E > x) \sim x^{-\gamma}, \quad \gamma < \infty]$$

Implication:

power-law tail in energy/fluence distributions

excess extreme events compared to Gaussian models

Prediction SP3: Correlated Transients

Test statistic:

$$[C(\tau) = E[S(t)S(t + \tau)]]$$

Null hypothesis:

$$[C(\tau) = 0]$$

HDSPH prediction:

$$[C(\tau) \neq 0 \quad \text{for structured subsets of events}]$$

Prediction SP4: Entropy Anomalies

Testable formulation:

$$[H_{obs}(S) < H_{null}(S)]$$

where:

(H_{null}) : entropy of best-fit noise model

(H_{obs}) : empirical signal entropy

Implication:

signals appear more structured than standard stochastic models predict

4.3 Falsification Criteria

Criterion F1: Null Model Sufficiency

HDSPH is rejected if:

$$[\forall D: \mathcal{L}HDS\mathcal{P}H(D) \leq \mathcal{L}null(D)]$$

i.e., no likelihood improvement over conventional astrophysical models.

Criterion F2: No Clustering Excess

If:

$$[N_{obs} \sim \text{Poisson}(\lambda)]$$

with no measurable overdispersion:

$$\left[\frac{\text{Var}(N)}{E[N]} = 1 \right]$$

then projection-based clustering is unsupported.

Criterion F3: Full Recoverability

If:

$$[I(M; O) \rightarrow H(O)]$$

then projection loss is unnecessary and HDSPH collapses to standard emission theory.

Criterion F4: No Residual Structure

If:

all residuals are white noise

no cross-event correlation exists

no heavy tails exist

then latent-field structure is unsupported.

Criterion F5: Model Non-Superiority

If simpler models achieve:

higher predictive accuracy

fewer free parameters

better Bayesian evidence

then HDSPH is rejected under Occam-consistent inference.

4.4 Future Observations

Observation Program O1: Large-Scale Transient Surveys

Data sources:

FRB catalogs

radio transient surveys

time-domain sky surveys

Target metrics:

clustering statistics

deviation from Poisson baseline

anisotropy in detection rates

Observation Program O2: Entropy-Based Analysis

Measured quantity:

$$[H(S)]$$

Goal:

Detect deviations:

$$[H_{obs} - H_{model} < 0]$$

Observation Program O3: Cross-Correlation Analysis

Statistic:

$$[C(\tau), \quad C_{ij}]$$

Goal:

Detect:

inter-event correlation

long-range temporal dependence

hidden coupling between catalogs

Observation Program O4: SETI Projection Search

Reformulated objective:

Search for:

statistically anomalous signals

non-stationary noise structures

low-probability high-information events

rather than assuming intentional encoding.

Observation Program O5: Multi-Observatory Coherence Tests

Hypothesis:

If latent field structure exists:

$$[S_1(t), S_2(t) \sim \text{correlated realizations of } M]$$

Test:

cross-site correlation above noise threshold

frequency-independent coherence patterns

synchronized anomaly detection

Final Scientific Statement (Revised)

HDSPH does not assert the existence of higher-dimensional or hidden physical structures.

Instead, it defines a **falsifiable statistical model class**:

$$[S_{obs} = \mathcal{H}(M) + \eta]$$

where deviations from standard stochastic emission models are encoded in:

clustering statistics

entropy reduction

cross-correlation structure

heavy-tailed event distributions

The hypothesis remains valid only if it demonstrates:

statistically significant improvement over null astrophysical models in explaining transient datasets.

Otherwise, it reduces to a non-superior descriptive framework.

Appendix A - Canonical Latent Model Classes

Purpose

Provide explicit prior model classes for the latent field (M) so forward modeling, inference, and hypothesis tests are well-posed and comparable.

Matérn Gaussian field

Definition: (M) is a Gaussian random field with Matérn covariance parameterized by range (ρ) and smoothness (ν).

Signature: smooth spatial correlations, finite correlation length, Gaussian marginal tails.

Heavy-tailed alpha-stable field

Definition: (M) has (α)-stable marginal distributions with ($0 < \alpha < 2$) and a specified dependence structure.

Signature: power-law tails, frequent extreme realizations, slow posterior concentration.

Nonstationary Gaussian process

Definition: (M) is a Gaussian process with spatially or temporally varying covariance parameters (for example, locally varying length scale).

Signature: localized bursts, changing correlation structure, operator-dependent detectability.

Cox process latent intensity

Definition: $\left(\lambda(x, t) = \exp\left(g(M(x, t))\right)\right)$ with (g) a link function and (M) a latent random field.

Signature: overdispersion relative to homogeneous Poisson, clustering driven by latent modulation.

Identifiability and Regularization

Identifiability statement

Recovery of (M) from observations (O) is identifiable only up to the null space ($\ker \mathcal{H}$). Practical identifiability requires priors or constraints that penalize components in ($\ker \mathcal{H}$) and stabilize inversion.

Regularization strategies

Tikhonov regularization

Minimize $(|\mathcal{H}M - O|_2^2 + \lambda|LM|_2^2)$ with (L) a differential operator and (λ) a regularization weight. Use for smoothness enforcement and numerical stability.

Sparsity priors

Impose (l_1) penalties or Laplace priors in a transform domain (wavelets, learned dictionary). Use to recover localized, transient structures and promote parsimonious explanations.

Hierarchical Bayesian priors

Place hyperpriors on covariance parameters or tail indices and infer jointly with (M). Use to quantify posterior uncertainty and propagate prior sensitivity into predictive checks.

Practical diagnostics

Report posterior concentration, effective rank of the Fisher information, and projection of posterior mass onto $(\ker \mathcal{H})$ to quantify irrecoverable modes.

Numerical Estimation and Inference Methods

Hierarchical Bayesian inference

Specify $(p(M, \theta))$ and likelihood $(p(O | M, \theta))$; sample the posterior with MCMC or HMC for low-to-moderate dimensional problems. Output: full posterior, credible intervals, posterior predictive checks.

Variational inference with mutual information lower bounds

Use variational approximations to obtain tractable lower bounds on $(I(M;O))$ and scalable posterior approximations for large problems. Output: mutual information lower bounds and fast approximate posteriors.

Penalized maximum likelihood and MAP

Optimize a penalized objective for large problems using conjugate gradients or proximal methods. Output: point estimates and Laplace approximations for uncertainty.

Tail estimation for heavy tails

Apply Hill estimator, Pickands estimator, or Bayesian tail models for fluence/energy distributions. Output: tail index $(\hat{\gamma})$ and confidence intervals.

Point process inference

Use Cox process or marked point process likelihoods with EM or MCMC for latent intensity estimation. Output: inferred $(\lambda(x,t))$, clustering diagnostics, and predictive event rates.

Practical Statistical Tests and Recommended Statistics

Prediction	Suggested test	Statistic
Heavy tails	Hill estimator; Pareto MLE	tail index $(\hat{\gamma})$
Non-uniform rates	Overdispersion test; likelihood ratio	Var/Mean ratio; LR
Event fragmentation	Ripley's K; pair correlation	$(K(r)); (g(r))$

Prediction	Suggested test	Statistic
Correlated transients	Autocovariance; coherence	$(C(\tau));$ coherence
Entropy anomalies	Permutation test; variational MI bound	$(H_{\text{obs}} - H_{\text{null}})$

Notes on implementation

Use bootstrap or Bayesian credible intervals for uncertainty; apply multiple-testing correction when scanning many regions or frequency bands; validate test power with forward simulations under candidate latent models.

Observational Requirements and Guidelines

Clustering and overdispersion

Minimum events: (≥ 100) independent detections per survey region to robustly reject homogeneous Poisson for moderate effect sizes.

Instrument requirement: stable detection threshold and well-characterized selection function.

Heavy-tail detection

Minimum events: (≥ 200) events to estimate tail index reliably with Hill estimator for moderate tails.

Dynamic range: at least two orders of magnitude in fluence or energy to distinguish power-law from exponential tails.

Mutual information and entropy tests

Minimum data: ensembles of ($>10^3$) independent signal windows improve estimator stability for high-dimensional features.

Preprocessing: consistent normalization, detrending, and noise whitening across datasets.

Cross-instrument coherence

Requirement: simultaneous or overlapping observations from at least two independent instruments with comparable sensitivity.

Minimum coincidences: tens of coincident detections to establish cross-site correlation above noise.

Practical note

These guidelines are heuristic and depend on effect sizes, noise levels, and model complexity. Perform power analysis with forward simulations under candidate latent models to refine sample size and sensitivity requirements for a given survey.

How to include this appendix in the manuscript

Place this material as **Appendix A** titled *Canonical Models, Identifiability, Inference, and Observational Guidelines*. Keep the appendix self-contained: include brief references to equations in the main text (for example, the forward model

$$S_{obs} = H[M] + \eta$$

and the intensity link

$$\lambda = G(M)$$

and add a short paragraph explaining that the canonical model classes and numerical recipes are recommended starting points for the analyses and simulations described in the main paper.

Appendix B — Cosmological Constants and Observational Fingerprints in HDSPH

Purpose of This Appendix

This appendix provides a comparative framework linking **standard cosmological constants and observational fingerprints** with their **HDSPH analogues**. The goal is to enable direct evaluation of whether projection-induced stochastic models can reproduce, distort, or reinterpret key astrophysical observables without invoking new physical emitters or exotic components.

B.1 Cosmological Constants and Their HDSPH Interpretations

B.1.1 Hubble Constant H_0

Standard interpretation: Expansion rate of the Universe inferred from the redshift–distance relation.

HDSPH interpretation:

$$H_{0,obs} = F_H(\rho_I, \lambda, W)$$

Apparent expansion may arise from:

- projection-induced **frequency drift**
- non-stationarity in the latent field
- anisotropic visibility due to the observation window

$$W(x, t)$$

Implication: HDSPH provides a natural mechanism for the **Hubble tension** as an operator-dependent distortion rather than a physical discrepancy.

B.1.2 Cosmological Constant $\Lambda\Lambda$

Standard interpretation: Dark energy density driving accelerated expansion.

HDSPH interpretation:

$$\Lambda_{\text{obs}} \propto \langle \rho_I(\mathbf{x}) \rangle H$$

A uniform component of the latent information density can mimic:

- apparent acceleration
- large-scale isotropy
- smooth background radiation

Implication: $\Lambda\Lambda$ becomes an **emergent projection effect**, not a fundamental energy component.

B.1.3 Matter Density Ω_m

Standard interpretation: Fraction of cosmic energy density in matter.

HDSPH interpretation:

$$\Omega_{\text{obs}} = \Phi(\lambda, \rho_I, K)$$

Clustering in the latent intensity field can reproduce:

- large-scale structure
- voids
- filamentary patterns

without requiring cold dark matter as a particle species.

B.1.4 Baryon Acoustic Oscillations (BAO)

Standard interpretation: Imprinted sound-horizon scale in matter distribution.

HDSPH interpretation:

$$\xi_{obs}(r) = \int K(r, x) \xi_M(x) dx$$

The BAO peak appears as a **projection-preserved correlation scale** in the latent field.

B.1.5 CMB Temperature and Spectrum

Standard interpretation: A 2.725 K blackbody relic radiation with small anisotropies.

HDSPH interpretation:

$$T_{obs} = \mathcal{H}(M_{bg}) + \text{noise}$$

smooth latent background field M_{bg} produces:

- near-perfect blackbody spectrum
- anisotropies via kernel convolution

- large-scale anomalies via windowing effects
-

B.2 Observational Fingerprints and HDSPH Counterparts

B.2.1 Redshift–Distance Relation

Standard fingerprint: Linear Hubble law at low redshift.

HDSPH counterpart: Frequency drift emerges from:

- latent field non-stationarity
 - projection-induced time dilation
 - spectral mixing through $K(y,x)K(y,x)$
-

B.2.2 Large-Scale Structure (LSS)

Standard fingerprint: Filaments, voids, clusters.

HDSPH counterpart:

$$LSS_{obs} = H(\rho_I) + \text{sampling noise}$$

Latent information density naturally forms:

- clustered regions
 - sparse voids
 - heavy-tailed mass distributions
-

B.2.3 CMB Anisotropies

Standard fingerprint: Multipole spectrum C_ℓ .

HDSPH counterpart:

$$C_{\ell, obs} = |K_\ell|^2 \times C_\ell(M) + N_\ell$$

Low- ℓ anomalies may arise from:

- anisotropic window functions
 - kernel asymmetries
 - partial sky sampling
-

B.2.4 FRB Dispersion Measures (DM)

Standard fingerprint: $DM \propto$ integrated electron density.

HDSPH counterpart:

$$DM_{obs} = H(\rho_{I^{ion}}) + \eta$$

Predictions:

- heavy-tailed DM distributions
 - clustering in DM–time space
 - instrument-dependent DM anomalies
-

B.2.5 Gravitational Wave Background

Standard fingerprint: Stochastic background with characteristic strain spectrum.

HDSPH counterpart:

$$H_{0,obs} = F_H(\rho_I, \lambda, W)$$

Latent fluctuations can mimic:

- stochastic backgrounds
- spectral breaks
- anisotropies

B.3 Comparative Table

Cosmological Fingerprint	Standard Interpretation	HDSPH Interpretation
Hubble constant H_0	Expansion rate	Projection-induced frequency drift
Λ	Dark energy	Mean latent information density
Ω_m	Matter fraction	Clustering of latent intensity
BAO	Early Universe sound waves	Preserved latent correlation scale
CMB	Relic radiation	Projected background field
LSS	Gravitational growth	Structure in ρ_I
FRB DM	Electron column density	Projected ionized latent field
GW background	Merging binaries	Latent stochastic fluctuations

B.4 Summary

This appendix establishes a direct mapping between **cosmological observables** and **HDSPH latent-field projections**, enabling:

- comparison with $\Lambda\Lambda\text{CDM}$
- identification of operator-induced distortions
- reinterpretation of anomalies (Hubble tension, CMB low- $\ell\ell$ anomalies, FRB DM outliers)
- formulation of testable predictions for cosmological datasets

HDSPH does not replace cosmology; it provides a **projection-based alternative lens** through which standard fingerprints can be re-evaluated.

Appendix C — Cosmological Constants Affecting Transient Signals

Comparative Table: Physical Fingerprints vs HDSPH Interpretations

Cosmological / Physical Constant	Standard Physical Effect on Signals	Observed Fingerprint (e.g., WOW!, Y-Signal)	HDSPH Interpretation (Projection-Based)
Hydrogen 21 cm line (1420.405751 MHz)	Fundamental hyperfine transition; reference for SETI	WOW! at 1420.4556 MHz ($\approx +50$ kHz offset)	Offset arises from kernel-induced spectral mixing or latent frequency drift , not necessarily Doppler or transmitter tuning
Cosmic Redshift z	Frequency shift due to expansion	Narrowband signals slightly offset from atomic lines	Apparent shift emerges from non-stationarity in $M(x,t)$ or projection-induced time dilation
Dispersion Measure (DM)	Integrated electron density along line of sight	Frequency-dependent delay; FRB-like dispersion	DM becomes $H(p_{\text{lion}}) \rightarrow$ heavy-tailed DM, anomalous DM–time clustering
Rotation Measure (RM)	Faraday rotation by magnetic fields	Polarization angle rotation	RM becomes kernel-dependent mixing $\rightarrow$ non-linear polarization drift without requiring strong B-fields
Scattering Time τ_{sc}	Multi-path propagation broadening	Exponential tail in time profile	Tail arises from convolution with $K(y,x)$ rather than ISM turbulence alone
Hubble Constant H_0	Cosmological expansion $\rightarrow$ redshift	Slight systematic drift in narrowband signals	Drift emerges from latent field evolution , not cosmic expansion
Cosmological Constant Λ	Accelerated expansion $\rightarrow$ large-scale isotropy	Smooth background noise floor	Background = latent information density floor projected through H

Cosmological / Physical Constant	Standard Physical Effect on Signals	Observed Fingerprint (e.g., WOW!, Y-Signal)	HDSPH Interpretation (Projection-Based)
Matter Density Ω_m	Structure formation → LSS	Clustering of transient sources	Clustering = inhomogeneous ρ_I , not necessarily matter distribution
Electron Density n_e	Dispersion, scattering	DM anomalies, scattering tails	Appears as projection of latent ionized component
Magnetic Field B	Faraday rotation	RM anomalies	RM anomalies = kernel asymmetry , not necessarily strong B-fields
Cosmic Microwave Background (CMB)	2.725 K blackbody background	Noise floor + anisotropies	Background = latent smooth field , anisotropies = windowing effects
Plasma Frequency f_p	Cutoff for propagation	Low-frequency suppression	Suppression = frequency-dependent kernel response
Refractive Index Variations	Frequency-dependent bending	Spectral wandering	Wandering = latent field fluctuations , not ISM lensing
Gravitational Potential Φ	Shapiro delay	Time-delay anomalies	Delay = non-local mixing in H

Special Row: Y-Signal–Type Events

Feature of Y-Signal	Standard Physical Interpretation	HDSPH Interpretation
Frequency near hydrogen line	Intentional tuning or astrophysical coincidence	Emergent from latent spectral density peaks + kernel convolution
Narrowband structure	Artificial transmitter or maser	Projection of a localized high-density region in M
Non-repetition	One-off emitter or noise	Sparse Poisson realization of latent intensity field
High SNR	Strong emitter	High local ρ_I crossing detection threshold

Feature of Y-Signal	Standard Physical Interpretation	HDSPH Interpretation
Lack of Doppler drift	Stationary source	Operator cancellation of latent drift components

Y-Signal Frequency Fingerprint Comparison

Quantity	Symbol / Unit	Observed / Assumed for Y-Signal	Standard Interpretation	HDSPH Interpretation
Central frequency	f_c [MHz]	$f_c \approx 1420.45$ (near 21 cm line)	Intentional tuning to hydrogen line or astrophysical coincidence	Peak of latent spectral density projected by H; proximity to 21 cm arises from spectral selection of the kernel, not necessarily intent
Offset from H line	$\Delta f = f_c - f_{21\text{cm}}$ [kHz]	$\Delta f \sim +40\text{--}60$ kHz	Doppler shift (relative motion), gravitational redshift, or instrumental offset	Effective offset from non-stationarity in $M(x,t)$ and frequency-dependent kernel response ; can be produced without bulk motion or fine tuning
Instantaneous bandwidth	Δf_{BW} [kHz]	Narrowband, $\Delta f_{\text{BW}} \ll f_c$	Coherent emitter (artificial or maser-like), high Q-factor source	Projection of a locally coherent patch of the latent field; narrowband arises from spectral filtering in H and selection effects
Signal-to-noise ratio	SNR [linear or dB]	High SNR single event	Strong emitter or favorable propagation path	High local information density p_I crossing detection threshold; a rare extreme of a heavy-tailed latent field, not necessarily a powerful source

Quantity	Symbol / Unit	Observed / Assumed for Y-Signal	Standard Interpretation	HDSPH Interpretation
Temporal duration	T [s]	$T \sim 10^1 - 10^2$ s (WOW!-like)	Source on-time, beam crossing, or scintillation	Length of time the observation window $W(t)$ overlaps a high-intensity region in $M(x,t)$; duration set by window $\times$ latent dynamics , not intrinsic lifetime
Frequency drift	$\dot{f}$ [Hz/s]	$\dot{f} \approx 0$ (no measurable drift)	Stationary source in barycentric frame or fine-tuned compensation	Cancellation of latent drift by the measurement operator: different drift components in M are mixed and averaged by H , yielding an apparently stable tone
Repetition	—	No confirmed repeats	One-off emitter, rare astrophysical event, or RFI	Sparse Poisson realization of a latent intensity field: low rate λ in the relevant region of $M \rightarrow$ non-repetition is expected, not exceptional
Spectral shape	$S(f)$	Single narrow peak over noise	Coherent carrier, engineered signal, or narrow astrophysical line	Projection of a localized mode in the latent spectral field; shape determined by kernel $K(f,x)$ and not uniquely diagnostic of artificiality
Polarization (if p measured)		Unknown / instrument-dependent	Constrains emission mechanism and propagation medium	In HDSPH, polarization is another channel of H ; complex or unstable polarization can arise from kernel anisotropy rather than exotic media

Interpretive summary: Under standard assumptions, a Y-signal–like event near the hydrogen line with high SNR, narrow bandwidth, and no drift is suggestive of a **coherent, possibly artificial emitter**. Under HDSPH, the same fingerprint is reinterpreted as a **rare, high-intensity projection** of a latent stochastic field, shaped by the measurement operator’s spectral kernel and observation window. The key claim is not that the signal is non-physical, but that **its apparent “designed” features can emerge from projection geometry and heavy-tailed latent structure** without invoking intentional transmission.